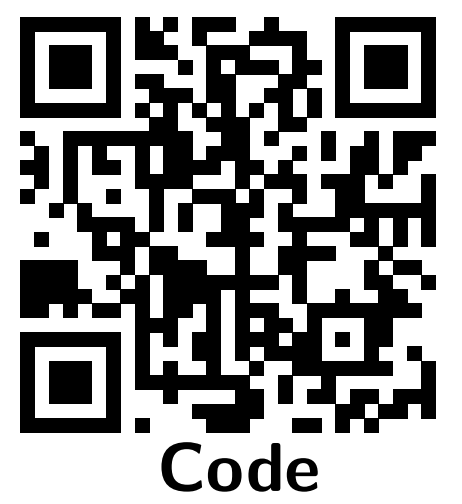


Inherently Interpretable Graph Neural Networks via B-cos Alignment

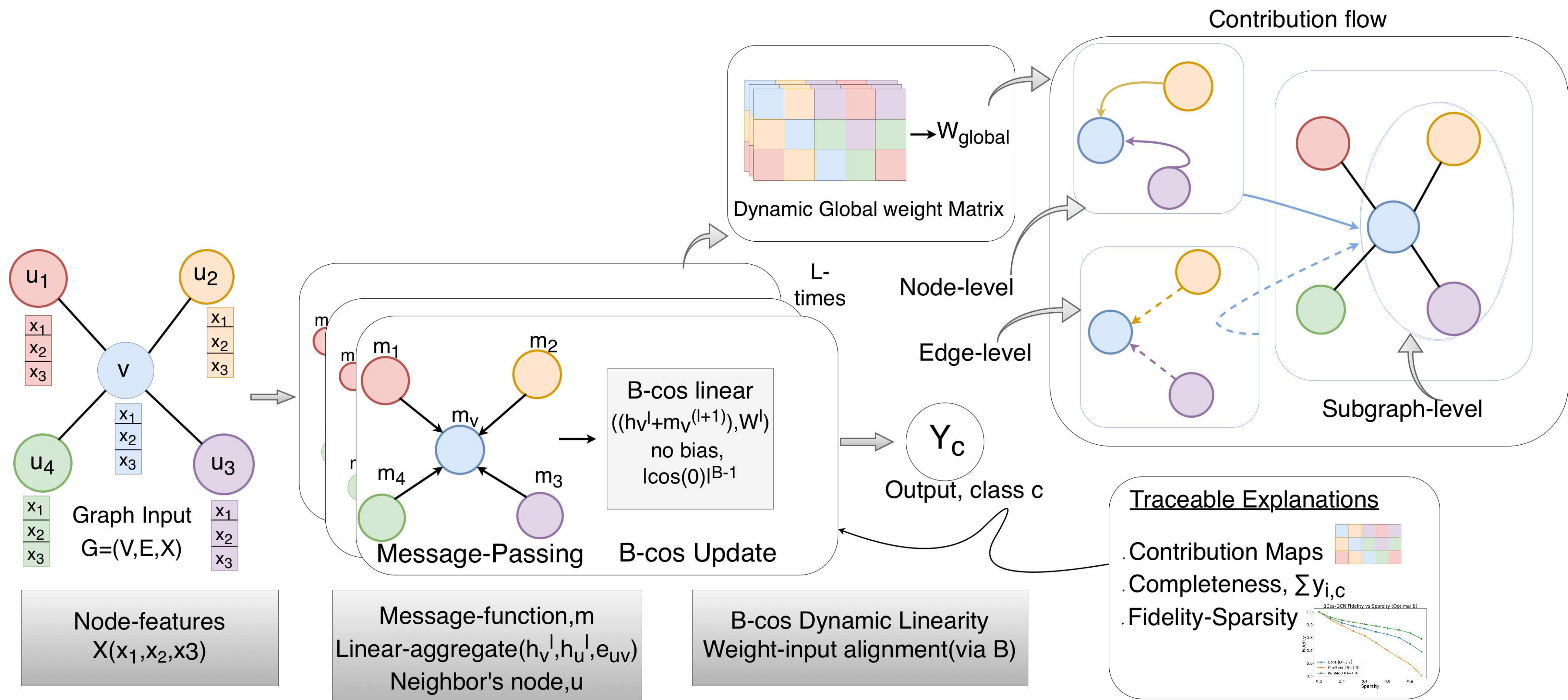
Interpretability is part of the architecture, not a separate explainer

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Code

Method Workflow



Graph Input → Message Aggregation → B-cos Update → Global Dynamic Weight Matrix → Contribution Maps

1. Aggregate. Each message-passing layer aggregates neighbour information exactly as in the baseline GNN.

2. Align. The standard linear update is replaced with a bias-free B-cos update that rewards weight-input alignment.

3. Decompose. Across layers the network acts as an input-dependent linear map $W_{1 \rightarrow L}(X, G)$, so the class score splits exactly into node, edge, and subgraph terms.

Motivation

Graph Neural Networks achieve strong predictive performance, but their decision process is hard to inspect. Post-hoc explainers produce useful visualisations, yet they approximate model behaviour after training and need not reflect the actual inference path.

Research Question

Can a GNN provide faithful explanations directly from its own forward computation while staying competitive in accuracy?

B-cos Transform

The B-cos operation rewards weight-input alignment. Features aligned with the learned direction are emphasised, weakly aligned components are suppressed.

$$\text{B-cos}(x, w) = \|w\| \|x\| |\cos \theta|^{B-1} \text{sign}(x^\top w)$$

$B \geq 1$ controls alignment strength.

No bias: contribution paths stay traceable.

Dynamic linearity: stacked layers still collapse into an input-dependent linear map.

Contributions

- ▶ Reformulate GNN message passing with B-cos transformations.
- ▶ Preserve input-output alignment across layers.
- ▶ Collapse the full network into a global dynamic-linear map.
- ▶ Evaluate accuracy, fidelity, sparsity, and explanation cost on seven benchmarks.

Built-in explanations: Node ✓ Edge ✓ Subgraph ✓

Explanation Output

Node contribution: influential nodes and node features.

Edge contribution: information flow between neighbours.

Subgraph contribution: compact explanatory subgraphs selected by node and edge contribution thresholds.

Attribution extraction: $\mathcal{O}(nd)$ per query

Theoretical Properties

Alignment behaviour: the B-cos form keeps a directional relation between input features and the learned representation across layers.

Dynamic-linear representation: for a fixed input graph the model is an input-dependent linear map

$$f(X) = W_{1 \rightarrow L}(X, G) \text{vec}(X)$$

Unified contribution analysis: the same map yields node-, edge-, and subgraph-level scores from one derivation.

Complexity: per-layer cost stays at the baseline GNN level

$$\mathcal{O}(ed + nh)$$

Scores come from the forward computation itself, so they track the prediction path without a separate post-hoc explainer.

Experimental Setup

Node classification: Cora, CiteSeer, PubMed, Texas, Cornell

Graph classification: MUTAG, PROTEINS

Backbones: GCN, GraphSAGE, GAT and their B-cos variants.

Metrics: accuracy, fidelity, sparsity, explanation time, memory.

Alignment exponent:

$$B \in \{1.0, 1.2, 1.3, 1.5, 1.7, 2.0, 2.5, 3.0\}$$

Predictive Performance

B-cos variants stay competitive with their baselines.

Model	Cora	CiteSeer	PubMed	Texas	Cornell	MUTAG	PROTEINS
GCN	0.79 ± 0.01	0.69 ± 0.01	0.77 ± 0.00	0.59 ± 0.00	0.51 ± 0.01	0.73 ± 0.10	0.70 ± 0.06
B-cos GCN	0.81 ± 0.01	0.70 ± 0.01	0.79 ± 0.01	0.69 ± 0.01	0.59 ± 0.04	0.73 ± 0.07	0.72 ± 0.05
GraphSAGE	0.80 ± 0.00	0.71 ± 0.00	0.76 ± 0.01	0.82 ± 0.01	0.75 ± 0.02	0.74 ± 0.06	0.69 ± 0.04
B-cos GraphSAGE	0.80 ± 0.01	0.70 ± 0.01	0.76 ± 0.00	0.86 ± 0.00	0.81 ± 0.02	0.75 ± 0.05	0.70 ± 0.04
GAT	0.80 ± 0.01	0.66 ± 0.01	0.77 ± 0.01	0.58 ± 0.01	0.49 ± 0.02	0.75 ± 0.11	0.69 ± 0.03
B-cos GAT	0.77 ± 0.01	0.64 ± 0.01	0.75 ± 0.01	0.66 ± 0.05	0.56 ± 0.05	0.75 ± 0.10	0.70 ± 0.04

Backbone-specific results on Cornell. B-cos improves both corresponding baselines: GCN increases from 0.51 ± 0.01 to 0.59 ± 0.04 , while GraphSAGE increases from 0.75 ± 0.02 to 0.81 ± 0.02 .

GAT observation. B-cos GAT shows a modest accuracy reduction ($\leq 2\%$) due to attention mechanism constraints.

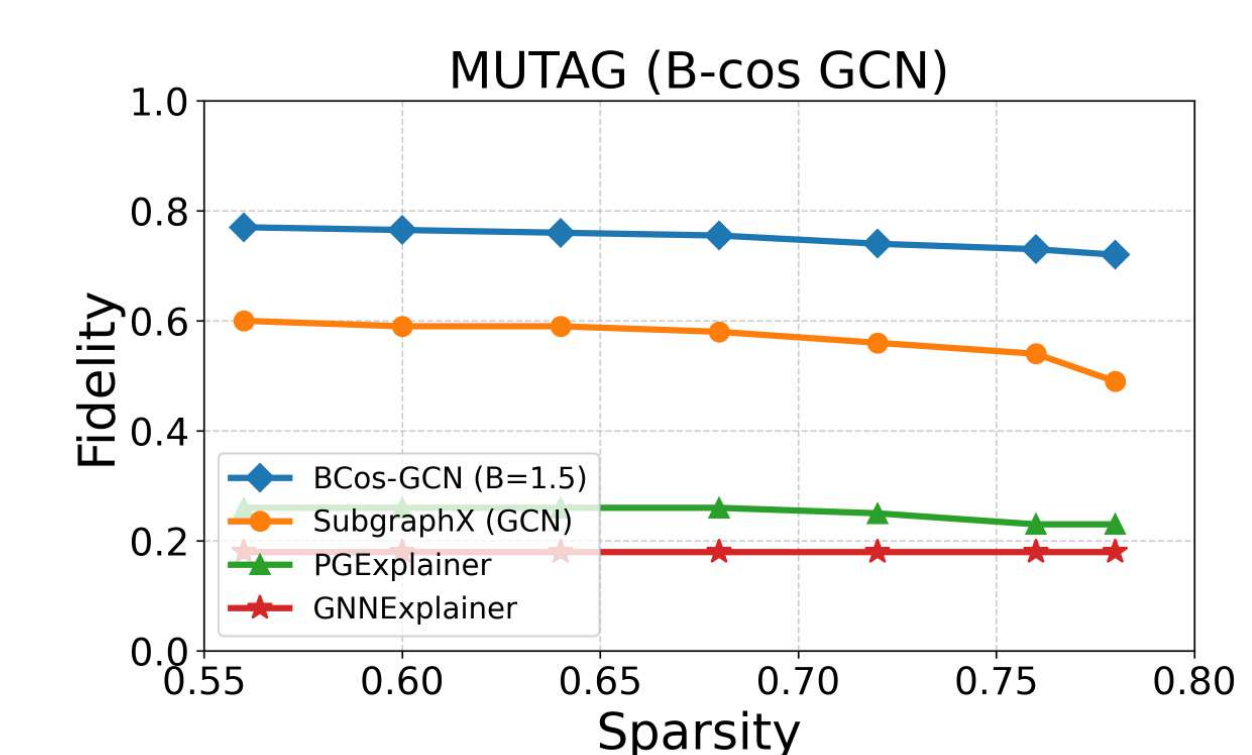
Explanation Cost

Attributions are read off the forward pass, so no optimisation loop is needed per explanation.

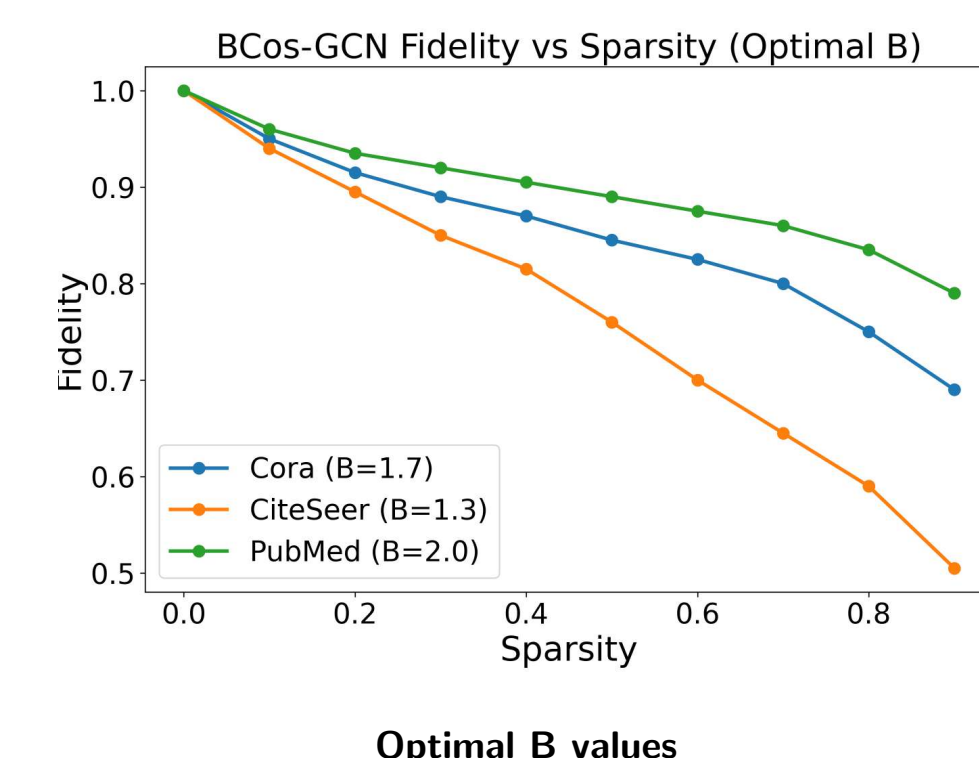
Method	Time (ms)	Memory (MB)
GNNE explainer	152 ± 12	45.2
SubgraphX	380 ± 25	45.2
B-cos GNN	0.8 ± 0.1	46.1

One forward pass, no per-instance search.

Fidelity and Sparsity



The fidelity-sparsity curves test how well the prediction survives when only a compact subset of high-contribution components is kept.



Across datasets, fidelity decreases monotonically with sparsity at the selected optimal B values, demonstrating stable and faithful explanation behavior.

Takeaways

- ▶ Faithfulness comes from the architecture, not from a surrogate fitted after training.
- ▶ Accuracy is retained, with clear gains on the heterophilous datasets.
- ▶ One exponent, B , trades alignment strength against fit.